

nology in village areas) creates an opportunity for drones to be used by the Panchayatiraj department in aerial land surveys for land demarcation in the rural areas in order to resolve land disputes faster. NITI Aayog's endeavour, DISHA (Drone for Infra Security and Agriculture) is being conceptualised to create an incubation fund for the drone industry. It aims to promote the R&D, bridging the gap between the academia and the industry, thus accelerating the growth of the ecosystem that is scalable and sustainable.

A counter-drone solution too is on the cards as reflected in the dialogues with the security agencies. BVLOS drones are already being planned to be flown, without having the UIN, on experimental basis by the consortia of thirteen operators, as approved by the MoCA, including Spice Jet, Nandan Nilekani backed ShopX, Google supported Dunzo, and drone maker Throttle Aerospace.

Though six green zones are already activated for drone flights, yellow zones too are going to be allowed, covering another 25-30% land.

In nutshell, a solid playground is being levelled to involve active participation of both public and private enterprises.

## Enterprises Taking the Lead

- Enterprises are proactively taking steps to create demand for commercial drones.
- Food tech company, Zomato, for example, is looking at setting up a drone-based delivery network in India.
- Tata Steel has deployed drones at their Noamundi iron ore mines in Jharkhand with the help of Skylark Drones, a Bengaluru-based data analytics startup, in order to meet the compliance of reporting and monitoring volumetric production.
- A bauxite company hired Skylark to leverage their drone-based monitoring solution to ensure safe deployment of all equipment and manpower at the mines.
- The drone technology developed by

## SVAMITVA Target

1. Short term target : Map 100,000 villages by Dec. 2020
2. Long term target – Map 660,000 (approx.) villages by Dec. 2024

Table E

Chennai-based DeTect Technologies acts like its own positioning system, more like an internal GPS helping in monitoring structures like a boiler inside a plant, autonomously. Hindustan Petroleum, Indian Oil, and GAIL have already roped in DeTect for developing similar solutions.

- Though still in the testing phase, IT-based businesses such as Amazon, Facebook, Google etc are slated to invest over \$200 million in the use of drones for delivery and expanding internet access.

The technology has to be plug & play. There has to be an industry forum where the industry aspirants should be able to get accurate information, do the verification of the suppliers and buyers. It should also provide a framework for funding to a spectrum of technologies. But it is imperative for the start-ups to develop a niche technology, before approaching the potential funding."

Organisation like FICCI or CII need to work as national level integrator and develop a set of database comprising R&D facilities, testing facilities, etc.

—Group Captain Jagtishwar Singh, NSCS

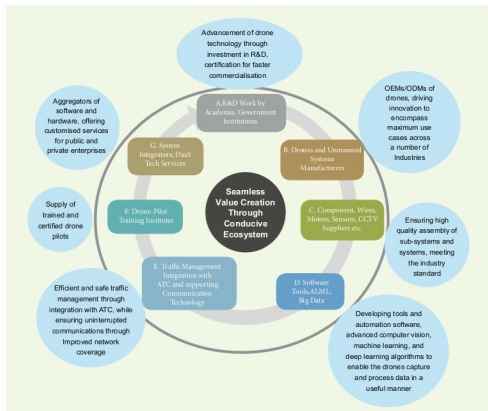


Fig. 5: Ecosystem of drones

## The academia to set the stage for innovation and advancement in drone technology

The academia is of the opinion that initially the funding should be routed through the academic institutions. There should be research grants from the government in order to make Indian drones competitive in the global market. IIT Kanpur and Indian Institute of Science are already providing mentorship and formal education to the entrepreneurs and the aspiring professionals.

The academia is going to play a pivotal role in commercialisation of drone technology. IIT Kanpur's test labs accommodate a series of technology focused sections including propulsion set up, environmental testing, communication testing, etc. The institute is willing to broaden its scope of the technologies as the industry matures and more and